Quantum verification of full-continuum Coulomb energy with unit nuclei

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Abstract

We prove QMA membership for the electronic Coulomb spectral-infimum problem with an explicitly listed set of distinct rational unit-charge nuclei, a unary number of spin-1/2 electrons, and inverse-input-length threshold separation. The minimization is over the full antisymmetric continuum space. No binding, attained ground state, nuclear separation, physical diameter, supplied orbital basis, or state-preparation promise is imposed. An IMS decomposition over all inner particle numbers, followed by exterior fermionic packets, gives one finite box even for unbound infima. Hardy estimates control a Coulomb cap, and bounded-potential spectral compression gives a physical momentum cutoff. Exact shifted sampling on the retained band supplies a uniform potential-form estimate without a configuration-space volume factor. Explicit symmetry and cutoff projections then give soundness for arbitrary quantum witnesses. Together with the stipulated unit-charge continuum hardness theorem of OpenAI, this proves QMA-completeness for the same promise problem.

1 The problem and antecedents

An input lists $M \geq 1$ pairwise distinct points $R_1, \ldots, R_M \in \mathbb{Q}^3$, an integer $N \geq 1$ in unary, and rationals $a < b$. Every coordinate and threshold is encoded as a signed binary numerator and positive binary denominator in lowest terms, with the fixed standard self-delimiting encoding of [1]. Let L be the total binary and unary input length, and require $b - a \geq L^{-1}$. Every nucleus has charge one. In atomic units the electronic Hamiltonian is

$$H = -\frac{1}{2}\sum_{i=1}^{N} \Delta_{x_i} - \sum_{i=1}^{N}\sum_{\alpha=1}^{M} \frac{1}{|x_i - R_\alpha|} + \sum_{1\leq i<j\leq N} \frac{1}{|x_i - x_j|}. \tag{1.1}$$

Its Hilbert space and form domain are

$$\mathcal{H}_N = \bigwedge^{N} (L^2(\mathbb{R}^3) \otimes \mathbb{C}^2),$$
$$\mathcal{Q}_N = H^1(\mathbb{R}^{3N}; (\mathbb{C}^2)^{\otimes N}) \cap \mathcal{H}_N.$$

Antisymmetry exchanges spatial and spin coordinates together. We use the self-adjoint operator of the closed, lower-bounded Coulomb form and set $E = \inf \operatorname{spec} H$. Nuclear repulsion is omitted. The YES promise is $E \leq a$; the NO promise is $E \geq b$. No requirement is imposed outside the promise.

Theorem 1. *The promise problem above belongs to QMA. The verification circuits and witnesses have size polynomial in L, uniformly over all allowed inputs.*

Corollary 2. *Assuming the unit-charge continuum hardness theorem [1, Theorem 1], the same promise problem is QMA-complete under deterministic classical polynomial-time many-one reductions.*

The source theorem is used for hardness only and is taken as a premise at repository revision `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`. Membership is proved below. Neither statement assumes neutrality, strict binding, an eigenvector at the infimum, an excitation gap, a selected spin sector, a magnetic field, confinement, or an orbital basis. Nuclear positions may be exponentially large or exponentially close, subject only to their binary descriptions and distinctness. A witness need not be efficiently preparable. The result does not include exponentially large binary nuclear charges; the companion hardness theorem [2] has that different input class.

Closest upper bounds. O'Gorman, Irani, Whitfield and Fefferman prove QMA-completeness with a supplied finite orbital basis [3]. Their published Appendix D, Corollary 1, adds positive unit nuclei while retaining that basis; Section V distinguishes the complete-space question. Matrix-element estimates inside that basis do not give the input-uniform continuum comparison needed here. Zheng, Leng, Liu and Wu prove StoqMA membership for smooth few-body potentials with polynomially bounded C^1 norm on Dirichlet boxes whose side lengths are polynomially bounded above and below [4, Theorem 3.3]. Their proof uses finite differences and a unary coordinate embedding. This pays the bounded smooth-box problem. Direct rescaling of an exponentially long box changes the derivative and kinetic scales, and their scalar minimum does not impose the fermionic sector.

Real-space simulation provides further antecedents. Kivlichan et al. control Fourier truncation and sampled norms, with general estimates containing a supremum-to-volume factor and a stronger derivative premise for their polynomial-spacing corollary [5, Theorem 4, Corollary 5, Lemmas 9–15]. Childs et al. use Fourier-tail estimates with a high-derivative parameter; their Coulomb discussion uses a periodic box and a softened interaction [6, Sections 2.2, 2.5 and Appendix B]. Schuch and Verstraete's Hubbard membership and designed-field realization are earlier complexity antecedents [7]. We do not substitute any of these dynamic or supplied-basis interfaces for a uniform spectral-infimum certificate.

The verification interface. Hardy inequalities, IMS localization, spectral compression, Fourier orthogonality, quantum Fourier transforms, and finite-group projections are standard tools. Sector-valued localization for correlated states is classical: Lewin's Proposition 3.1 and Example 3.3 explicitly construct localized Fock states by partial traces and retain every particle-number sector [8]. The ionization companion [9] credits that interface; its strict-binding and radius results are not used here. Our contribution is the complete uniform conversion from the singular, unconfined fermionic input to a finite-register verifier, with both energy inequalities and all-witness soundness. The comparison is bounded by these inspected primary interfaces, not a worldwide-priority assertion.

2 Uniform form estimates and the energy range

Throughout, norms and integrals sum over spin indices. Put

$$T[\psi] = \frac{1}{2}\sum_i \|\nabla_i \psi\|_2^2, \qquad C = 4NM^2, \qquad g = b - a, \qquad \eta = \min\{1, g/100\}, \qquad D = 3N. \tag{2.1}$$

We have $M, N \le L$ and $\eta^{-1} \le 100L$.

Three-dimensional Hardy, with the other coordinates fixed, gives

$$\int \frac{|\psi|^2}{|x_i - y|^2} \le 4 \, \|\nabla_i \psi\|_2^2 \,. \tag{2.2}$$

The center y may be a nucleus or another particle coordinate. Integrating all remaining variables preserves the constant. For normalized ψ, Cauchy–Schwarz and Young's inequality imply

$$\sum_{i,\alpha} \int \frac{|\psi|^2}{|x_i - R_\alpha|} \le 2M \sum_i \|\nabla_i \psi\|_2 \le 2\sqrt{2} M \sqrt{N} \sqrt{T[\psi]} \le \tfrac{1}{2} T[\psi] + C. \tag{2.3}$$

Repulsion is nonnegative, so the Coulomb form q satisfies

$$q[\psi] \ge \tfrac{1}{2} T[\psi] - C \, \|\psi\|_2^2 \,. \tag{2.4}$$

The version of Young's inequality with an arbitrary positive kinetic coefficient proves infinitesimal form bounds for all the finitely many Coulomb terms. It justifies the closed-form convention in Section 1. The same estimates hold on Dirichlet open subsets by zero extension.

In particular $E \ge -C$. To see $E \le 0$, choose N normalized smooth packets of width R with disjoint supports separated by at least R^2. Their normalized wedge, with any chosen spin labels, has kinetic energy $O(N/R^2)$ and repulsion $O(N^2/R^2)$. Attraction is nonpositive. Letting $R \to \infty$ proves

$$-C \le E \le 0. \tag{2.5}$$

No minimizing vector is required.

For $\varepsilon > 0$ set $v_\varepsilon(r) = \min\{r^{-1}, \varepsilon^{-1}\}$, including $v_\varepsilon(0) = \varepsilon^{-1}$, and cap every attractive and repulsive kernel in q to obtain q_ε. Since $0 \le r^{-1} - v_\varepsilon(r) \le \varepsilon / r^2$, Hardy gives

$$|q_\varepsilon[\psi] - q[\psi]| \le 8\varepsilon (M + N) T[\psi]. \tag{2.6}$$

For a pair $i < j$, use the i coordinate in (2.2); each coordinate is counted at most $N - 1$ times. The cap reduces attraction and keeps repulsion nonnegative, so (2.4) also holds for q_ε. No separation parameter enters either estimate.

3 One finite box, including escaping electrons

Define

$$A_0 = 1 + \max_\alpha \|R_\alpha\|_\infty \,, \quad R_0 = 2A_0, \quad s = \left\lceil \frac{100(NM + N^2 + N + 1)}{\eta} \right\rceil . \tag{3.1}$$

All nuclei lie in the ball of radius R_0. Choose a Lipschitz function $\theta(r)$ equal to 0 for $r \le R_0 + s$ and to $\pi/2$ for $r \ge R_0 + 2s$, with $|\theta'| \le 2/s$. Let $\chi_{\text{in}}(x) = \cos\theta(|x|)$ and $\chi_{\text{out}}(x) = \sin\theta(|x|)$. Their squares sum to one and their squared gradients sum to at most $4/s^2$.

For $S \subseteq \{1, \dots, N\}$ put

$$\psi_S = \prod_{i \in S} \chi_{\text{in}}(x_i) \prod_{i \notin S} \chi_{\text{out}}(x_i) \psi.$$

The product partition has $\sum_S \|\psi_S\|_2^2 = \|\psi\|_2^2$. The IMS identity, with kinetic coefficient $1/2$, yields

$$q[\psi] \ge \sum_S q[\psi_S] - \frac{2N}{s^2} \, \|\psi\|_2^2 \,. \tag{3.2}$$

Lipschitz cutoffs suffice for this form identity; they can also be smoothly approximated while retaining the bounds. This use of all sectors is the classical localization mechanism discussed in [8].

Let E_k^{ball} be the k-electron Coulomb Dirichlet infimum, with the same nuclei, in $B(0, R_0 + 2s)$, and set $E_0^{\mathrm{ball}} = 0$. For a fixed S, fixing the outside positions and spins leaves, for almost every slice, an H_0^1 spinor antisymmetric among the $|S|$ inside particles. The common inside cutoff preserves this antisymmetry and vanishes at the ball boundary. After normalizing a nonzero slice, its internal form is at least $E_{|S|}^{\mathrm{ball}}$; multiplying by its squared norm and integrating gives the same bound for the original unnormalized slice. Full antisymmetry across an inside and an outside label is not needed in this step.

Every outside electron is at distance at least s from every nucleus. Discarding its nonnegative kinetic energy and all repulsions involving it therefore gives

$$q[\psi_S] \geq \left(E_{|S|}^{\mathrm{ball}} - NM/s\right) \|\psi_S\|_2^2 .$$

Combining this with (3.2) and taking arbitrary near-infima proves

$$\min_{0 \leq k \leq N} E_k^{\mathrm{ball}} \leq E + \frac{NM}{s} + \frac{2N}{s^2}. \tag{3.3}$$

In particular the argument retains the vacuum sector and unbound states.

Conversely fix k and a normalized near-minimizer for E_k^{ball}. Add $N - k$ packets supported in cubes of side s, centered at

$$c_j = (R_0 + 2s + (4j + 2)s, 0, 0), \qquad 1 \leq j \leq N - k.$$

Each packet is the product of the three normalized first Dirichlet sine modes, extended by zero, with spin up. Its kinetic energy is $3\pi^2/(2s^2) < 15/s^2$. All new supports are separated from the inner ball and from one another by at least s.

Wedge these packets with the arbitrary correlated inner k-particle state. This is an isometry because their one-particle support spaces are orthogonal. More explicitly, expansion of the antisymmetrization gives disjoint assignments of coordinate labels to support regions; their normalization factors cancel. The same expansion preserves all internal energy expectations. Every new repulsive pair costs at most $1/s$, and there are at most $N^2/2$ of them. Added attraction is nonpositive.

Let B be the least power of two no smaller than $\lceil R_0 + (4N + 6)s \rceil$, and set $Q = (-B, B)^3$. Every support just used lies inside Q. If E_Q is the N-particle Dirichlet infimum in Q, zero extension and the preceding construction give

$$E \leq E_Q \leq \min_k E_k^{\mathrm{ball}} + \frac{15N}{s^2} + \frac{N^2}{2s} \leq E + \eta. \tag{3.4}$$

Indeed $NM/s + N^2/(2s) + 17N/s^2 < \eta$ by (3.1). No algorithm must identify the minimizing k: it proves existence of a witness with exactly N particles. Both $k = 0$ and $k = N$ are legal. Although B can be exponential, $\log B = O(L + \log s + \log N)$.

4 Capping and physical momentum compression

On Q choose

$$\varepsilon = \frac{\eta}{128(M + N)(C + 2)}, \qquad \alpha = 8\varepsilon(M + N) = \frac{\eta}{16(C + 2)}. \tag{4.1}$$

Write E_ε for the capped antisymmetric Dirichlet infimum. An uncapped near-minimizer has $T \le 2(C + \eta + o(1))$ by (2.4) and (3.4). Its capped energy thus proves $-C \le E_\varepsilon \le E_Q + \eta/8 \le 2$. Conversely a capped near-minimizer has $T \le 2(C + 2 + o(1))$. Using (2.6) in the reverse direction and taking limits yields, with slack,

$$|E_\varepsilon - E_Q| \le \eta/4. \tag{4.2}$$

This explicitly pays for capping repulsion as well as attraction.

The capped multiplication potential V_ε satisfies

$$\|V_\varepsilon\|_\infty \le B_V := \frac{N(M+N)}{\varepsilon}, \qquad \mathrm{Lip}(V_\varepsilon) \le G := \frac{2N(M+N)}{\varepsilon^2}, \qquad J = C + 2 + B_V. \tag{4.3}$$

A nuclear term has Lipschitz constant at most ε^{-2} and a pair term at most $\sqrt{2}\varepsilon^{-2}$ in all D coordinates; summing proves the bound. It is independent of nuclear clustering.

The Dirichlet kinetic operator T_Q has compact resolvent. A bounded self-adjoint potential preserves its operator domain and compact resolvent. Particle permutations commute with both terms, so this remains true on the antisymmetric subspace. There is a normalized ground vector ψ for the capped operator, in $\mathrm{Dom}(T_Q)$, with

$$\|T_Q\psi\|_2 \le |E_\varepsilon| + B_V \le J. \tag{4.4}$$

No analytic regularity of an uncapped Coulomb eigenfunction is assumed.

Set $\ell = 2B$, translate the cube to $(0,\ell)^3$, and choose

$$P = \lceil 64J/\eta \rceil, \qquad K = \ell P. \tag{4.5}$$

Let F keep the product sine modes with $1 \le n_j \le K$ in every scalar coordinate, with all spin indices retained. The same cutoff in every particle coordinate commutes with permutations. For $p = F\psi$ and $r = (1-F)\psi$, an omitted mode has kinetic eigenvalue at least $P^2/2$. Equation (4.4) therefore implies

$$\|r\|_2^2 \le \frac{4J^2}{P^4}, \qquad \langle r, T_Q r\rangle \le \frac{2J^2}{P^2}. \tag{4.6}$$

Since $(H_\varepsilon - E_\varepsilon)\psi = 0$,

$$\langle p, (H_\varepsilon - E_\varepsilon)p\rangle = \langle r, (H_\varepsilon - E_\varepsilon)r\rangle \le \frac{2J^2}{P^2} + \frac{4J^3}{P^4}.$$

Here $J \ge 1$, $P^2 \ge 8J$, $\|p\|_2^2 \ge 3/4$, and $4J^2/P^2 \le \eta/8$. Normalizing p and using the variational lower bound for every retained vector gives the compressed minimum E_F:

$$E_\varepsilon \le E_F \le E_\varepsilon + \eta/8. \tag{4.7}$$

The absolute scalar physical frequencies are below $W = 4P$, since $\pi < 4$. Thus the retained kinetic norm is at most $T_{\max} = 8DP^2$. All quantities in (4.3) and $P, W, T_{\max}$ are polynomial-valued in L. The integer K need not be, but its binary length is polynomial.

5 Sampling with a uniform form bound

Oddly extend each sine coordinate from $(0,\ell)$ to the circle of length 2ℓ. Multiplication by $2^{-D/2}$ makes the resulting extension an L^2 isometry. On the torus, represent each coordinate by $t \in [-\ell, \ell]$, fold it to $|t| \in [0,\ell]$, translate by $-B$, and evaluate the original capped potential. Call this even

extension V^{ext}. The fold is 1-Lipschitz for circle distance, including its joins; hence $\|V^{\mathrm{ext}}\|_\infty \le B_V$ and $\mathrm{Lip}(V^{\mathrm{ext}}) \le G$. Kinetic and potential forms agree under the normalized odd extension. All extensions commute with particle permutations.

Choose q to be the least power of two satisfying

$$q > 4K, \qquad h := 2\ell/q \le \frac{\eta}{16D(G + 2B_V W + 1)}. \tag{5.1}$$

This choice is computable by rational comparisons. It has polynomial binary length. We use the grid $x_j = 2\ell j/q$, interpreted modulo 2ℓ, for $j \in \{0, \dots, q-1\}^D$.

For explicit normalization, the orthonormal torus modes, indexed by $n \in \mathbb{Z}^D$, are

$$e_n(t) = (2\ell)^{-D/2} \exp(\pi i n \cdot t/\ell).$$

A bandlimited spinor is $f = \sum_{n \in [-K,K]^D} c_n e_n$, with vector-valued spin coefficients. Define the sampling vector by

$$(S_u f)_j = h^{D/2} f(x_j + u).$$

Every frequency difference lies in $[-2K, 2K]^D$, and $2K < q$. The identity $\sum_{j=0}^{q-1} e^{2\pi i m j/q} = 0$ for nonzero $|m| < q$ therefore proves, for every shift u,

$$\|S_u f\|_2^2 = \|f\|_2^2 = \sum_n \|c_n\|_{\mathbb{C}^{2^N}}^2 . \tag{5.2}$$

The phase of a shift does not change this calculation. It also proves inner-product preservation, including for translated differences. There is no dimension-dependent approximation factor.

For normalized bandlimited f, integration of its directional derivative along a translation gives

$$\|f(\cdot + u) - f\|_2 \le |u| \, \|\nabla f\|_2 \le W\sqrt{D}\, |u|.$$

For $u \in [0, h)^D$, (5.2) gives $\|S_u f - S_0 f\|_2 \le WDh$. Set

$$A_u(f) = h^D \sum_j V^{\mathrm{ext}}(x_j + u) |f(x_j + u)|^2,$$

with spin summation understood. Both sample vectors have norm one. The elementary inequality $\sum_j ||v_j|^2 - |w_j|^2| \le 2\,\|v - w\|_2$ for unit vectors, and the Lipschitz bound for the potential, imply

$$|A_u(f) - A_0(f)| \le G\sqrt{D}\, h + 2B_V W D h. \tag{5.3}$$

Averaging A_u over one grid cell reproduces the exact torus integral, by partition into translated cells. From (5.1),

$$\left| A_0(f) - \left\langle f, V^{\mathrm{ext}} f \right\rangle \right| \le \eta/16 \tag{5.4}$$

for every normalized bandlimited state. By homogeneity and the Hermitian spectral characterization this is an operator-norm bound on the compressed potential; it holds for mixed states as well. It has no wavefunction supremum norm, high derivative, or factor involving ℓ^D.

The q-point discrete Fourier transform identifies S_0 with position and momentum registers. It preserves the kinetic values $\frac{1}{2}(\pi/\ell)^2 \sum_j n_j^2$. Let $\mathcal{A}$ be the joint subspace of signed frequencies $|n_j| \le K$, odd parity in each scalar coordinate, and full particle antisymmetry. Reflection of a position register is $j \mapsto -j \pmod q$. Oddness removes zero modes; the cutoff excludes the Nyquist frequency. Thus $\mathcal{A}$ is exactly the sampled Dirichlet antisymmetric sine space, with no additional boundary modes.

Let E_{disc} be the minimum on $\mathcal{A}$ of the compressed sum of that kinetic diagonal and the sampled potential diagonal. Equations (3.4), (4.2), (4.7), and (5.4) give

$$|E_{\mathrm{disc}} - E| \le \eta + \eta/4 + \eta/8 + \eta/16 < 2\eta. \tag{5.5}$$

The lower inequality applies to every valid witness. The potential need not preserve the cutoff: only its compressed quadratic form is measured.

6 A uniform validity instrument

Each of the D spatial registers has $\log_2 q$ qubits, and there are N spin qubits. The following ideal projections have uniform polynomial-size approximations in a fixed universal gate set.

First apply the inverse Fourier transform, interpret each index as a signed residue in $[-q/2, q/2)$, and compare its absolute value with K. Reject on failure; uncompute the comparison and return to position representation. Next measure every reflection involution $j \mapsto -j \pmod q$ by a controlled-reflection test, accepting only its -1 eigenspace. The cutoff and these odd projectors commute. Their full product commutes with particle permutations because each particle has the same three restrictions. Individual odd projectors need not commute with those permutations.

To impose fermionic symmetry, use digits $d_i \in \{1, \dots, i\}$ for $i = 1, \dots, N$. A permutation is uniquely specified by inserting label i at position d_i in the previously ordered list of $1, \dots, i-1$. Uniform independent digits therefore give a uniform permutation. Starting from the ordered register blocks, the corresponding permutation can be implemented using at most $N(N-1)/2$ controlled adjacent swaps: at stage i, move block i left to slot d_i. A padded sequence tests, for each possible swap position, whether it must be performed. Each test is a comparison on $O(\log N)$ bits. The parity is $\sum_i (i - d_i) \pmod 2$, computable by reversible arithmetic. If one chooses the inverse convention for the resulting permutation action, inversion preserves both the uniform distribution and the sign. Each swapped block contains its three spatial registers and its spin qubit.

Prepare uniform digit states, compute the swap controls and sign, perform the controlled block permutation and sign phase, then uncompute auxiliary classical data and reverse the digit preparation. Accept only if the digit ancilla returns to its initial zero state. The accepted operator is

$$\mathsf{A} = \frac{1}{N!} \sum_{\pi \in S_N} \operatorname{sgn}(\pi) U_\pi. \tag{6.1}$$

Inversion makes it self-adjoint; multiplying the group sums proves $\mathsf{A}^2 = \mathsf{A}$. Its range is exactly $U_\pi \psi = \operatorname{sgn}(\pi)\psi$ for every π. For $N = 1$ it is the identity. No $N!$-term enumeration is performed.

Since (6.1) commutes with the full previous projection, the accepted Kraus operator is precisely $\Pi_{\mathcal{A}}$. Any input density operator ρ therefore leaves the subnormalized state

$$\Pi_{\mathcal{A}} \rho \Pi_{\mathcal{A}} \tag{6.2}$$

on success; failure is rejection. This is an actual instrument, not a postselection promise about the witness.

For clarity on uniform gate complexity, a uniform interval of size i can be prepared through a depth-$O(\log i)$ binary decision tree. Given a prefix, reversible arithmetic counts the allowed integers in its two children and computes the required square-root rotation angle. Counts are computed from the prefix rather than by enumerating all nodes. Computing an angle to t bits and implementing rotations controlled on those bits costs polynomial time in $\log i$ and t; rounding errors can be made inverse polynomial. Reversing the circuit gives its inverse. Approximate Fourier transforms,

comparisons, controlled permutations and fixed-gate synthesis also have polynomial cost at inverse-polynomial operator accuracy. A telescoping bound over the polynomial number of stages controls the acceptance error uniformly over all witnesses, *before* conditioning. Small success probabilities cannot amplify that error.

7 Arithmetic, energy testing and amplification

The grid coordinates have polynomial binary length although their values and the grid size may be exponential. Folding and translating require rational arithmetic of polynomial bit cost. For each of the $O(NM + N^2)$ kernels, compare the exact squared distance with ε^2. Below that threshold, including exact coincidence, return ε^{-1}. Otherwise compute the inverse square root to the needed absolute accuracy. One may bisect its value in $[0, \varepsilon^{-1}]$, testing a candidate using rational squared comparisons. The number and bit cost of bisection steps are polynomial. Summing at sufficiently small per-term error evaluates the potential uniformly over all grid points.

Kinetic values are also computable in polynomial time: indices have polynomial bit length, π^2 is computable to the needed precision, and admitted physical frequencies are bounded by W. Off the accepted band one may clamp the kinetic value to $[0, T_{\max}]$; this keeps approximate instruments bounded and does not change their action on $\mathcal{A}$. There is no need to enumerate grid points or evaluate an uncapped collision.

We first remove extreme thresholds using (2.5). Accept if $b > 0$ or $a \geq 0$, when a NO instance is impossible. Otherwise reject if $a < -C$ or $b \leq -C$, when a YES instance is impossible. Outside the promise any outcome is permitted. In the remaining case

$$-C \leq a < b \leq 0. \tag{7.1}$$

Set

$$t = (a + b)/2, \qquad Q_* = T_{\max} + B_V + C + 1, \qquad \gamma = g/(10Q_*). \tag{7.2}$$

After the validity instrument, choose fairly between two diagonal tests. For a Fourier kinetic outcome x, accept with probability $1/2 - (x - t)/(2Q_*)$. For a position potential outcome y, accept with probability $1/2 - y/(2Q_*)$. Both probabilities belong to $[0, 1]$. On a normalized valid state ρ, their average is

$$\frac{1}{2} - \frac{\operatorname{tr}(\rho H_{\mathrm{disc}}) - t}{4Q_*}. \tag{7.3}$$

The computed probabilities can be implemented by a random-bit register and comparator, or by a synthesized rotation. All arithmetic, gate and measurement-probability errors can together be made at most $\gamma/10$ in acceptance, uniformly over inputs. This has polynomial cost: after (7.1), $g \leq C$, all normalization constants are polynomial-valued, and $g \geq L^{-1}$. Thus γ^{-1} is polynomial.

For a YES input, (5.5) supplies a valid witness of energy at most $a + 2\eta$. The ideal validity test accepts it surely, and its ideal acceptance is at least

$$\frac{1}{2} + \frac{g/2 - 2\eta}{4Q_*} \geq \frac{1}{2} + \gamma,$$

since $\eta \leq g/100$ gives a margin at least $0.12g/Q_*$. For a NO input every normalized valid state has energy at least $b - 2\eta$, so its ideal acceptance is at most $1/2 - \gamma$. For an arbitrary witness, (6.2) multiplies such a nonnegative acceptance probability by its surviving weight, at most one. Hence the same soundness bound holds. Circuit approximations leave margins $1/2 + 0.9\gamma$ and $1/2 - 0.9\gamma$.

Repeat on an odd number r of registers, taking r to be the least odd integer at least $10\gamma^{-2}$, and accept a majority. Tensor copies of a good witness give completeness at least $2/3$ by the Bernoulli tail bound. For soundness, let B_{acc} be the one-register acceptance POVM effect. On a NO instance, $\|B_{\mathrm{acc}}\| \le 1/2 - 0.9\gamma$. The majority effect is a sum of tensor products of B_{acc} and $I - B_{\mathrm{acc}}$. Their product eigenbasis reduces each eigenvalue to a Bernoulli majority probability with each success parameter at most that bound. Monotonicity and the same tail estimate bound the largest eigenvalue by $1/3$. This applies to entangled witnesses across the registers; no independence promise is imposed on Merlin. The witness size and uniform circuit construction remain polynomial in L. This proves Theorem 1, and [1, Theorem 1] gives Corollary 2.

8 Scope, provenance and formal support

The cap bound treats arbitrarily close nuclei. The binary grid handles exponentially distant ones. The all-k localization and exterior wedge include ions, escaping electrons and unattained infima. Every spin sector is retained. The uniform form estimate and the actual projections prevent invalid high-frequency, aliased or wrong-statistics witnesses from evading soundness. These are analytic and circuit arguments, not numerical tests. The polynomial total charge is essential to the estimates and follows from the explicit unit-nucleus list; no binary-large-charge extension or efficient witness preparation follows.

The hardness premise is the exact theorem at the pinned revision stated in Section 1; it is not re-proved here. The original scope and the earlier feasibility sketch were recorded separately before completion of this proof. Independent internal proof and contribution reviews accepted the unchanged scope; these are not external peer review or a worldwide-priority certificate. The supporting record gives source hashes, actual primary reading limits, reproducible build instructions and the exact correspondence with the reviewed proof.

Scoped formal support. The companion Lean development proves normalized finite-character Parseval identities for an injective family of additive characters, preservation under specified unit-modulus coefficient phases, and the translation identity on a finite abelian group. It specializes to characters of $\mathbb{Z}/q\mathbb{Z}$ and proves injectivity of the centered residue modes under $2K + 1 \le q$. Separately, for a supplied complex representation of a finite group, it proves the formula, idempotence, character-eigenvector range and commutation properties of the character-weighted average, including the permutation-sign specialization. Scalar lemmas prove the total error bound from the four assumed intermediate energy inequalities, the fair-test identity, quantitative YES/NO margins, and soundness after a subnormalized weight and an assumed acceptance approximation error.

These are supporting finite-algebraic and scalar statements. They do not formalize continuous torus integration or the production of phases by real translations, the product-grid and spin realization, the concrete particle representation, or Hilbert-space self-adjointness and orthogonality of the average. Hardy/IMS localization, the exterior wedge, the Coulomb cap, operator-domain and Ritz estimates, the Lipschitz quadrature argument, quantum instruments and their polynomial circuit approximations, and entangled-witness amplification remain in the human proof above. In particular the assumed scalar energy and acceptance bounds are not Lean proofs of their analytic or circuit hypotheses. Exact declarations, compiler/dependency versions, source hashes, controls and coverage limits are recorded in the accompanying support files; no complete formalization of Theorem 1 is claimed.

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